

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our analytical model demonstrates a strong capability in predicting outcomes, accurately classifying approximately 83% of records. This robust performance indicates that the insights derived from this analysis are highly reliable, providing a solid foundation for strategic decision-making. The model significantly outperforms previous analytical approaches, offering a clearer understanding of the factors influencing outcomes. The primary drivers influencing these outcomes have been identified as gender, economic status (represented by the fare paid), and age. Specifically, the male demographic shows a significant influence on predictions, while the financial commitment (fare) and age demographics also play a crucial role. Understanding these core demographic and economic characteristics of our customer base is critical for targeted initiatives. Given the strong predictive power and the clear identification of these key factors, we recommend a focused approach on monitoring trends within these specific customer segments. Changes in the profiles of these influential groups—particularly gender, economic background, and age—are likely to have the most significant impact on overall business outcomes. Regularly tracking these variables will enable proactive adjustments to strategies, ensuring we remain responsive to market dynamics and customer behavior.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

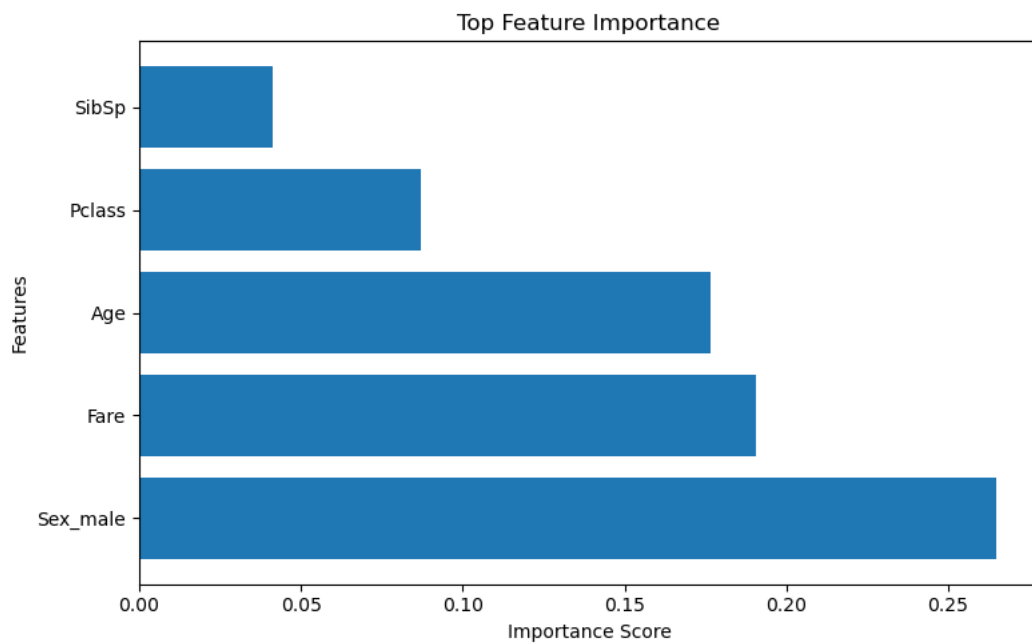
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

